

EC-341 Digital System Design

BS(CE)-2021

LAB REPORT # 08



Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.
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Experiment #08

Implementation of Verilog Parameters

Objective

- **Verilog Parameters**
- **Examples**
- **Lab Tasks**

Software Used

- Vivado

Task:

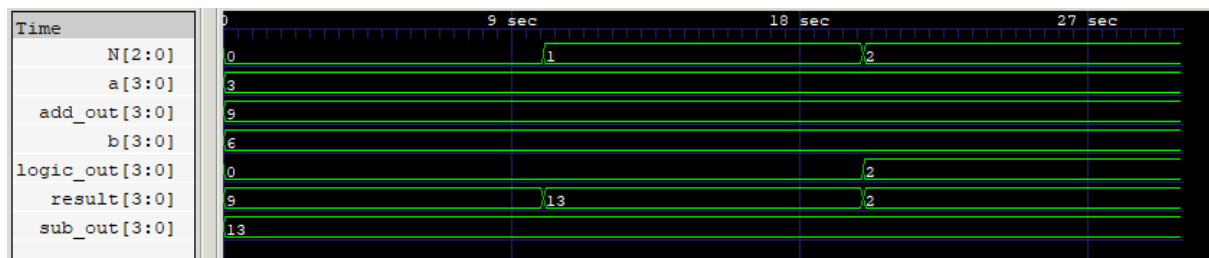
Code:

```

1  module parameterized_alu(
2      input wire [3:0] a,
3      input wire [3:0] b,
4      input wire [2:0] N,
5      output reg [3:0] result
6  );
7
8      wire [3:0] add_out;
9      wire [3:0] sub_out;
10     wire [3:0] logic_out;
11
12     assign add_out = a + b;
13     assign sub_out = a - b;
14     assign logic_out = (N == 2) ? (a & b) : (N == 3) ? (a | b) : (N == 4) ? (a ^ b) : 4'b0;
15
16     always @(*)
17     begin
18         case(N)
19             3'b000: result = add_out;
20             3'b001: result = sub_out;
21             3'b010, 3'b011, 3'b100: result = logic_out;
22             default: result = 4'b0;
23         endcase
24     end
25
26 endmodule
27
28 module testbench;
29
30     reg [3:0] a;
31     reg [3:0] b;
32     reg [2:0] N;
33     wire [3:0] result;
34
35     parameterized_alu alu(
36         .a(a),
37         .b(b),
38         .N(N),
39         .result(result)
40     );
41
42     initial begin
43         $dumpfile("dump.vcd");
44         $dumpvars(1,alu);
45         a = 4'b0011;
46         b = 4'b0110;
47         N = 3'b000;
48         #10 N = 3'b001;
49         #10 N = 3'b010;
50
51         #10;
52     end
53 endmodule

```

Output:



Conclusion:

In this lab session, we have implemented a simple parameterized ALU with 4 bit operands and 3 bit operations.